
Amitesh Gautam

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Summary

Skilled Blockchain Developer with 1 year of open source experience in designing, developing, and deploying decentralized applications (dApps) on Ethereum. Proficient in Solidity and ethers.js, with a strong understanding of smart contract development, blockchain architecture, and Ethereum standards like ERC-20. Adept at creating secure, gas-optimized smart contracts and integrating dApps with frontend frameworks like React. Experienced in using development tools such as Hardhat and Remix IDE to streamline testing and deployment. Passionate about leveraging blockchain technology to create scalable, secure, and innovative decentralized solutions.

Experience

- Independent Project Work** July 2023 - Present
(Blockchain Developer)
Developed and Deployed multiple Solidity-based smart contracts for diverse blockchain applications, including Decentralized Finance (DeFi), NFTs, and DApps, using industry-standard tools such as Hardhat and Remix IDE.
Designed and Implemented ERC-20 and NFT token standards to enable custom token creation, trading functionalities, and interactive user experiences within Ethereum-based ecosystems.
Enhanced User Interactivity in DApps with React.js and ethers.js, enabling seamless connections to Ethereum blockchain for transactions and real-time data display.
Architected Secure Multi-Signature Wallets and advanced DeFi protocols, leveraging Solidity to ensure high security, transparency, and decentralized governance.
Conducted Thorough Testing of smart contracts to ensure security and functionality using Hardhat, achieving robust, bug-free deployments.
Deployed and Interacted with Contracts on test networks, demonstrating practical knowledge of Ethereum blockchain architecture and smart contract lifecycle management.

Projects

- (Multi-Sigs) Multi-Signature Wallet**
A Multi-Signature Wallet (or "Multi-Sig Wallet") is a smart contract-based wallet that requires multiple authorized signatures to approve and execute transactions. Unlike traditional wallets, which rely on a single private key, a multi-signature wallet enhances security by requiring a predefined number of approvals from a group of trusted parties before funds can be accessed or transactions are finalized. This wallet is ideal for organizations, partnerships, or any setup that demands collaborative fund management. Key features typically include adding owners, storing ERC-20 tokens, setting signature requirements, viewing transaction history, and executing secure, multi-signature transactions.
- Decentralized Escrow Application**
Developed a Decentralized Escrow Application on the Ethereum blockchain to facilitate secure transactions between buyers and sellers without the need for intermediaries. The application leverages smart contracts written in Solidity to ensure that funds are only released when both parties meet predefined conditions.
- ECDSA Node**
This project begins with a client that is allowed to transfer any funds from any account to another account. That's not very secure. By applying digital signatures we can require that only the user with the appropriate private key can create a signature that will allow them to move funds from one account to the other. Then, the server can verify the signature to move funds from one account to another.
 1. Incorporated Public Key Cryptography so transfers can only be completed with a valid signature
 2. The person sending the transaction should have to verify that they own the private key corresponding to the address that is sending funds
- ETH Block Explorer**
This project is a responsive Ethereum Block Explorer built using React.js, Vite, and ethers.js. The app allows users to view the latest 18 Ethereum blocks, inspect block details, and drill down into transaction details within a selected block.
- Sepolia Faucet**
This project implements a simple Ethereum Sepolia Faucet contract using Solidity and Hardhat. The Faucet allows users to withdraw a limited amount of ETH at once and ensures that the contract is funded upon deployment. It includes automated tests for contract functionality and integrates Hardhat Ignition for deployment management.
- EIP-1559 ETH Gas Estimator**
A simple JavaScript-based gas fee estimator for Ethereum transactions using the EIP-1559 model. This tool calculates the estimated gas fees based on user inputs for the base fee, priority fee (tip), and gas limit.
- ETH Donation Split**
A smart contract to split incoming donations into two equal half for two recipients.

Education

- **Alchemy University**
Ethereum Bootcamp 2024 | Jun - Nov
- **Udemy**
Blockchain A-Z: Build a Blockchain, a Crypto + ChatGPT Prize 2023-24
- **Delhi University**
BA in History 2020-23

Skills

- Solidity | JavaScript | React | Node.js | Express.js | MongoDB | AWS | GraphQL Ethers | Hardhat | Remix IDE
- Cryptography | Blockchain Architecture | AI-Blockchain Integration | Query Blockchain
- Smart Contracts - Testing | Debugging | Gas Optimization | Deployment | Interaction | Upgrading
- ERC-20 | Non-Fungible Tokens (NFT) | Multi-Sigs - Multi Signature Wallet | Smart Contract Libraries | Proxy Contracts | Decentralized Applications (DApps) | Decentralized Finance (DeFi)
- API | (Model-View-Controller)MVC | Socket.io | AWS Lambda | Serverless Framework | Nginx | Mongoose ODM | Postman API | Rest APIs |